



MARCO ANTONIO MELO

SOFTWARE ENGINEER | CLOUD ARCHITECT

PROFILE

Full Stack Engineer with extensive experience designing and scaling distributed web platforms using the MERN stack and Python. Proven track record of leading complex system architecture initiatives, driving cloud-native solutions on Google Cloud Platform, and delivering highly available, secure, and cost-efficient services. **Google Cloud Certified Professional Architect**, with deep expertise in microservices, event-driven architectures, CI/CD, and infrastructure as code. Adept at cross-functional leadership, mentoring engineers, and translating business requirements into resilient technical solutions at scale.

WORK EXPERIENCE

IBM

2023 - PRESENT

Application Developer

Full Stack Engineer who led the maintenance, evolution, and governance of a companywide React and Tailwind design system, operating at the intersection of frontend architecture, backend developer platforms, and organizational standards. Drove multi-quarter initiatives that established enforceable governance, enabled safe large-scale migrations, and scaled consistent UI and architectural patterns across dozens of repositories and teams.

Contributions:

- **Design System Governance & Organizational Leadership:** Led the governance of a React + Tailwind design system adopted across 20+ repositories and 10+ product teams, defining API contracts, contribution standards, and usage guidelines that reduced UI fragmentation and improved cross-team consistency.
- **TypeScript Platform Enablement:** Led the rollout of full TypeScript support across the component library, contributing to an estimated 40–60% reduction in component-related runtime issues and significantly improving developer confidence and onboarding speed.
- **Automated Standards Enforcement (Custom ESLint Plugin):** Designed and implemented a custom ESLint plugin encoding design system and architectural rules, integrated into CI/CD pipelines to validate every change. This reduced design system violations by 80%+ within the first months and shifted compliance from manual reviews to automated enforcement at scale.
- **Backend Developer Platform (MCP Server):** Architected and built an MCP Server capable of parsing and reasoning over internal design system semantics, component APIs, and code patterns, powering intelligent tooling and validation across the ecosystem. This platform served as the foundation for advanced developer tooling and future automation initiatives.
- **Large-Scale Codemod Automation:** Developed AST-based codemod scripts to standardize component usage and architectural patterns across tens of thousands of lines of code, enabling safe, repeatable migrations and reducing manual refactor effort by 80%+ across the organization.

CONTACT

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EDUCATION

2017 - 2022

UVM

- Bachelor's in Computer Systems Engineering.

TECHNOLOGIES

- React
- React Native
- Expo
- Redux
- TypeScript
- JavaScript (ES6+)
- HTML5
- CSS3
- Tailwind CSS
- Chart.js
- Node.js
- Express
- Nodemailer
- Winston
- Pino
- REST
- CRUD
- JWT
- SQL Server
- SQL
- NoSQL
- Git
- Lerna
- ESLint
- AST
- Codemods
- MCP Server
- Python
- Java
- C#
- CI/CD
- Terraform
- Google Cloud Platform (GCP)
- Compute Engine
- GKE
- Cloud Run
- Cloud Storage
- Cloud SQL
- Firestore
- BigQuery
- Cloud Build
- Cloud Monitoring
- Cloud Logging
- Kubernetes
- Linux
- Windows Server
- Active Directory
- TCP/IP
- DNS
- XSS
- CSRF
- SSRF
- Vitest
- Jest
- Copilot

LANGUAGES

- English (Fluent) - TOEFL ITP.
- Spanish (Native).

KEY SKILLS

- **Frontend & Design Systems:** React, React Native, Redux, Tailwind CSS, TypeScript, JavaScript, Design Systems, Component Governance, Accessibility Standards.
- **Backend & Platform Engineering:** Node.js, REST APIs, GraphQL, Backend Platform Services, API Design, Data Modeling, Input Validation, Error Handling, Logging.
- **Developer Tooling & Automation:** ESLint, Custom ESLint Plugins, AST Parsing, Codemods, MCP Server, Developer Tooling Platforms, Monorepos (Lerna), Git.
- **CI/CD & Engineering Productivity:** CI/CD Pipelines, Automated Code Quality Gates, Static Analysis, Organizational Standards Enforcement.
- **Cloud Architecture (Google Cloud Platform):** Google Cloud Platform (GCP), Cloud-Native Architecture, Microservices, Load Balancing, IAM, VPC Networking, Cloud Storage, Managed Databases, High Availability & Scalability, Cost Optimization.
- **Security & Secure Engineering:** Authentication & Authorization, JWT, Cryptography, Secure Coding Practices, Web Security (XSS, CSRF, SSRF), Infrastructure Security, Threat Modeling.
- **Observability & Reliability:** Logging, Monitoring, Alerting, Reliability Engineering, Fault Tolerance.
- **Data & Persistence:** PL/SQL, NoSQL (MongoDB), Managed Databases, Data Integrity, Secure Data Access.
- **Cross-Platform & Mobile:** React Native, Expo, Mobile Security Best Practices
- **Systems & Networking Fundamentals:** Linux, TCP/IP, DNS, HTTP/HTTPS, Network Infrastructure Concepts.
- **Architecture & Leadership Practices:** System Design, Technical Governance, Platform Strategy, Developer Experience (DX), Cross-Team Enablement, Mentorship.

Deloitte Cyber Risk Advisor

2021 - 2023

Penetration Tester with experience conducting offensive security assessments across **banking, retail, energy, insurance, and government** environments. Combined real-world attack simulation with internal lab development to continuously sharpen exploitation techniques and validate detection and remediation strategies. Specialized in infrastructure, web, and binary exploitation, delivering actionable risk assessments grounded in realistic threat models.

Contributions:

- Performed infrastructure **penetration testing** across on-premises environments, identifying network segmentation weaknesses, insecure services, and critical misconfigurations in high-risk sectors.
- Conducted **web application penetration testing** against internal and external systems, uncovering vulnerabilities including authentication flaws, authorization bypasses, injection attacks, and business logic issues.
- Executed **binary exploitation and reverse engineering**, identifying memory corruption vulnerabilities and insecure custom implementations.
- Led **passive and active reconnaissance** using OSINT, service discovery, and attack surface mapping to emulate real threat actor methodologies.
- Achieved **privilege escalation, lateral movement, and pivoting** across segmented networks, demonstrating end-to-end compromise paths and real business impact.
- Designed and developed **multiple intentionally vulnerable applications** for internal security laboratories to support continuous training and tool validation:
 - A **Vanilla JavaScript web** application with controlled XSS and CSRF vulnerabilities.
 - A **Node.js REST API** exposing **SSRF**, authentication, and authorization flaws.
 - A **React application** intentionally vulnerable to DOM-based XSS and **JavaScript prototype pollution**.
- Used these labs to practice exploitation techniques, validate detection tooling, and improve exploit reliability in controlled environments.

Consorcio Jurídico de Cobranza Especializada 2020 - 2021 Mobile Developer

Mobile / Full Stack Developer with end-to-end ownership of a cross-platform mobile application built with React Native and Expo, backed by a secure Node.js REST API and a SQL Server data layer. Experienced in designing scalable mobile and backend architectures, applying security best practices, and delivering production-ready systems with a strong focus on reliability, maintainability, and performance.

Contributions:

- Architected and developed a cross-platform mobile application using React Native (Expo), ensuring high code quality, performance, and a consistent user experience across iOS and Android.
- Designed and implemented a Node.js REST API to support mobile workflows, encapsulating business logic, data access, and integrations.
- Built a mail service within the backend to handle user notifications, account lifecycle events, and transactional communications.
- Implemented JWT-based authentication and authorization, defining secure token lifecycles and access control mechanisms between the mobile client and backend services.
- Applied cryptographic best practices using industry-standard libraries to protect sensitive data, credentials, and secrets across the stack.
- Integrated and optimized a SQL Server database, designing schemas, indexes, and queries to ensure data integrity, scalability, and performance.
- Enforced security and architectural best practices across both mobile and backend layers, including secure storage, input validation, error handling, API hardening, and protection against common mobile and web vulnerabilities.

TECHNICAL EXPERTISE

Frontend & Platform Engineering

- Architect and scale complex frontend ecosystems using React, React Native (Expo), Redux
- Advanced TypeScript-first development (strong typing, generics, module augmentation, AST tooling)
- Design Systems architecture & cross-product component standardization
- Frontend architecture strategy (state orchestration, performance optimization, rendering patterns)
- Monorepo architecture (Lerna) & shared package governance
- Developer Experience (DX) optimization, ESLint customization, codemods, AST parsing
- Data visualization systems (Chart.js), mathematical graphing & equation rendering
- LMS platform architecture & large-scale UI systems

Backend & API Architecture

- Node.js service architecture & modular backend design
- REST API design at scale (versioning, backward compatibility, schema evolution)
- Data modeling & persistence strategy (SQL Server, SQL, NoSQL)
- Secure authentication & authorization architecture (JWT, RBAC patterns)
- Input validation pipelines & centralized error handling
- Structured logging & traceability architecture (Pino, Winston)
- Correlated request tracing & log standardization across distributed services
- Observability-aligned logging strategies (structured JSON logs, log levels, contextual enrichment)
- Mail services integration & transactional systems design
- Backend platform services & internal tooling systems

Testing Strategy & Quality Engineering

- Unit testing & integration testing architecture using Vitest and Jest
- Test-driven development (TDD) practices
- Component testing & API contract validation
- Mocking strategies & dependency isolation
- CI-integrated test automation pipelines
- Coverage enforcement & quality gates
- Testing standards governance across monorepos

Cloud & Distributed Systems Architecture (Google Cloud Platform)

- End-to-end GCP solution architecture design
- VPC topology design, subnet strategy & firewall governance
- IAM architecture & identity federation
- Compute Engine, GKE (Kubernetes Engine), Cloud Run architecture
- Cloud SQL, Firestore, BigQuery data architecture
- Infrastructure as Code (Terraform)
- CI/CD with Cloud Build
- Observability architecture (Monitoring, Logging, tracing strategy)
- Zero Trust security principles & cloud security hardening
- High availability & disaster recovery design
- Cost governance & FinOps strategy
- Hybrid & multi-cloud integration patterns

DevOps & Platform Reliability

- CI/CD pipeline architecture & deployment automation
- Git workflow governance & release management
- Secure configuration analysis
- Platform reliability & operational excellence
- Internal developer platforms & tooling systems
- MCP Server architecture & backend infrastructure services

Systems, Infrastructure & Enterprise Environments

- Network infrastructure fundamentals (TCP/IP, DNS)
- Active Directory & Windows Server environments
- Linux systems administration
- Enterprise authentication systems
- Web applications infrastructure & distributed system awareness

Application & Offensive Security Expertise

- Web application security (XSS, CSRF, SSRF, Prototype Pollution)
- Binary exploitation & reverse engineering fundamentals
- Assembly & memory corruption analysis
- Passive & active reconnaissance (OSINT)
- Service enumeration & vulnerability exploitation
- Privilege escalation, credential harvesting, lateral movement
- Post-exploitation analysis & pivoting
- Threat modeling & risk assessment
- Security reporting & remediation planning

- Led technical decision-making, aligned API contracts between frontend and backend, and ensured long-term maintainability through clean architecture and clear documentation.

LearnAla

2020 - 2020

Front End Developer

Front End Developer with hands-on experience building interactive and data-driven user interfaces for a Learning Management System (LMS) using React. Focused on translating complex mathematical concepts into intuitive visual experiences through dynamic charts, equation rendering, and responsive UI components.

Contributions:

- Developed and maintained React-based user interfaces and reusable components for an LMS platform, delivering interactive educational tools for math-focused learning experiences.
- Implemented Redux for predictable state management, coordinating application state across lessons, exercises, and visualization components.
- Integrated LaTeX rendering to accurately display mathematical notation and formulas, ensuring clarity and correctness in educational content.
- Built dynamic data visualizations using Chart.js, enabling real-time rendering of mathematical functions and learning outcomes.
- Designed and implemented frontend logic for graphing mathematical equations, including linear, quadratic, cubic, and higher-order functions, supporting dynamic inputs and real-time updates.
- Collaborated with educators and backend developers to align UI behavior with pedagogical goals and LMS content workflows.
- Focused on performance, usability, and accessibility to ensure a smooth learning experience across devices and user skill levels.

COURSES AND CERTIFICATIONS

- Test Driven Development with Angular [2025 updated] - Udemy.
- JavaScript (Intermediate) - HackerRank.
- JavaScript (Basic) - HackerRank.
- C# (Basic) - HackerRank.
- Java (Basic) - HackerRank.
- Python (Basic) - HackerRank.
- IBM WatsonX Essentials - IBM.
- IBM Consulting Way Habits (Foundational) - IBM.
- Enterprise Design Thinking Practitioner - IBM.
- Google Cloud Certified: Professional Cloud Architect - Google.
- Google Cloud Certified: Associate Cloud Engineer - Google.
- Google Cloud Certified: Cloud Digital Leader - Google.
- Retail Industry Jumpstart - IBM.
- Linux Commands: From Zero to Shell Script - Udemy.
- Malware Development Course 2022: From Zero to Hero - Udemy.
- React (Basic) Certificate - Hacker Rank.
- CCNA: Enterprise Networking, Security and Automation - Cisco.
- Angular: De cero a experto (Angular 10+) - Udemy.
- Vue.js [Actualizado] !De 0 a Experto! + Firebase + MEVN - Udemy.
- Complete React Developer in 2021 (w/ Redux, Hooks, GraphQL) - Udemy.